Rational Functions: Intercepts, Asymptotes, & Sign Charts

1. Anatomy of a Rational Function

A **rational function** is a quotient of two polynomials:

$$r(x) = \frac{p(x)}{q(x)} \quad \text{where } q(x) \neq 0$$

- **x-Intercepts (Roots):** Occur where the numerator vanishes, $p(a) = 0$ (with $q(a) \neq 0$). The point is $(a, 0)$.
- **Vertical Asymptotes (VA):** Occur where the denominator vanishes, $q(b) = 0$ (with $p(b) \neq 0$). The line is $x = b$.
- **y-Intercept:** Evaluated at $x = 0$: $r(0) = \frac{p(0)}{q(0)}$. The point is $(0, r(0))$.
- **Horizontal Asymptotes (HA):** Determined by degrees of $p(x)$ and $q(x)$ as $x \to \pm\infty$. When degrees are equal, $y = \frac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$.

2. Classroom Blackboard Example: $r(x) = \frac{(2x-1)(x+1)}{(x-1)(x+2)}$

- **x-Intercepts:** Set numerator to zero:

$$(2x-1)(x+1) = 0 \implies \mathbf{x = \frac{1}{2}}, \quad \mathbf{x = -1}$$

- **Vertical Asymptotes:** Set denominator to zero:

$$(x-1)(x+2) = 0 \implies \mathbf{x = 1}, \quad \mathbf{x = -2}$$

- **y-Intercept:**

$$r(0) = \frac{(2(0)-1)(0+1)}{(0-1)(0+2)} = \frac{(-1)(1)}{(-1)(2)} = \mathbf{\frac{1}{2}} \implies \left(0, \frac{1}{2}\right)$$

- **Horizontal Asymptote:** $\lim_{x\to\pm\infty} \frac{2x^2+x-1}{x^2+x-2} = \frac{2}{1} = \mathbf{2} \implies \mathbf{y = 2}$.

3. The Factor Sign Analysis Table

The critical values $\{-2, -1, 1/2, 1\}$ partition the real line into five intervals:

Interval	$(2x-1)$	$(x+1)$	$(x-1)$	$(x+2)$	**Total Sign of $r(x)$**
$x < -2$	$-$	$-$	$-$	$-$	$+$ (above axis)
$-2 < x < -1$	$-$	$-$	$-$	$+$	$-$ (below axis)
$-1 < x < 0.5$	$-$	$+$	$-$	$+$	$+$ (above axis)
$0.5 < x < 1$	$+$	$+$	$-$	$+$	$-$ (below axis)
$x > 1$	$+$	$+$	$+$	$+$	$+$ (above axis)

Sign Chart Rule:
The sign of $r(x)$ can only change at an x-intercept (crossing zero) or a vertical asymptote (blowing up across $\pm\infty$). Testing the sign of each linear factor within each partition completely determines where the rational curve lies above or below the x-axis.